GRADIA RESEARCH

Conditionally Approved: Proof-Bound Branchable Universes for Long-Horizon AI Agents Under Changing Evidence, Authority, and Time

Rudy M. Celekli, Gradia Research

Completed instrument study and assisted-attribution stress test - human adjudication pending - 25 August 2026

Contents

Completed-instrument boundary. The cost-capped execution and four-judge assisted-attribution stress test are complete. The execution aggregate was independently recomputed from 55 immutable attempt artifacts. This manuscript reports physical execution, exact machine-scored outcomes, infrastructure exclusions, runtime cohorts, cost, and trajectory shape. It does **not** claim a balanced model comparison, broad capability, frontier difficulty, tail reliability, human agreement, mortgage-domain validity, causal effects, training utility, or novelty. The 37 gradable outcomes remain behaviorally unattributed until blinded human adjudication distinguishes model failures from measurement defects.

Abstract

We study a measurement problem that static benchmarks cannot represent: an agent acts while its world changes, yet an evaluator must still prove what changed, when the agent could observe it, which authority controlled it, and whether later work used the current state. **Conditionally Approved** is a fully synthetic mortgage testbed for that apparatus. Its five long-horizon task editions isolate document truth under pushback, epistemic residue after retraction, competing legitimate authority, temporal portfolio control, and an honest handoff whose quality is measured by successor performance. Mortgage is the demanding test domain; the primary object of study is the proof-bound, evolving, branchable agent Universe.

The benchmark binds every eligible episode to a frozen task edition, model identity, runtime commit, action ledger, material-world roots, agent-visible projections, event and restore lineage, exact evaluator edition, and infrastructure disposition. Official reward is a strict binary conjunction over task correctness and evidence obligations; diagnostic vectors are reported separately and never converted into unregistered partial reward. We initially preregistered five attempts for each of four identity-pinned model providers. After attempt artifacts and zero outcomes already existed and had been inspected, the operator froze an operational cost-stop amendment: a hard ceiling of 55 physical attempts and a target of two gradable attempts per provider $\times$ task cell. This is explicitly **post-observation exploratory evidence**, not untouched preregistration. The selection rule remains outcome-blind within a cell—the lowest two gradable ordinals—and uniform zero rewards prevent the stop from selecting successes. The amendment preserves every attempt, excludes infrastructure failures from behavioral denominators, and forbids pass-at-five or tail-reliability claims.

The completed execution recorded 55 physical attempts across four identity-pinned providers: 37 were gradable and 18 were infrastructure exclusions. None of the 37 gradable attempts satisfied the exact perfect-rubric conjunction. Six of 20 provider-by-task cells contained an eligible same-runtime pair; neither attempt passed in any of those six pairs. Nine cells had only one gradable attempt and five were unobserved. The physical-attempt ledger recorded 3,617 provider calls, 99.38 million input tokens, 6.08 million output tokens, and \$705.88 in provider spend. The 37 gradable episodes—not the excluded partial traces—contained 2,609 official tool actions and 5,325 official transcript turns. A deterministic, non-reward engagement descriptor marked 13/37 gradable traces, including 7/21 selected primary attempts, as low engagement because they used under ten percent of the scripted control's actions and observed no source, attachment, or world event. Because execution crossed three runtime cohorts and did not cover all cells, the verifier refused a balanced pass-at-two statistic, provider ranking, and every pass-at-five claim. Behavioral attribution remains reserved for blinded human review. The benchmark was built and verified with the Gradia Universes platform; Gradia is the execution and evidence system, not the benchmark name.

A post-hoc assisted-attribution stress test then asked four independently pinned model judges to classify all 889 red surfaces from identical frozen evidence. Only 232 assignments (26.1%) were unanimous; descriptive Fleiss' kappa was 0.151, pairwise exact-label agreement ranged from 35.4% to 69.5%, and pairwise cited-anchor Jaccard ranged from 0.089 to 0.292. No machine judgment changed a frozen score. This is evidence that one confident judge is an unstable attribution authority, not evidence that plurality labels are human truth.

1. Why conditional approval is the right evaluation problem

An agent can produce a plausible final answer and still fail the work. It may cite a superseded source, silently obey an unauthorized request, miss a deadline boundary, carry retracted evidence into later decisions, or hand off a summary that causes the next agent to violate an active constraint. Conversely, a long or inefficient trajectory is not automatically incorrect. Conditioned on valid infrastructure, the official score asks whether the submitted work is correct and fully evidence-grounded under the frozen contract—not whether the agent used a preferred number of steps.

The title has a deliberate double meaning. In mortgage operations, approval is often conditional on unresolved evidence. In this benchmark, every evaluation claim is also conditional on a valid world, an admitted evaluator, and reviewable evidence.

Figure 1: The attribution chain evaluated in this paper. A changed world must be materially applied, disclosed to the agent under valid authority, and reflected in correct downstream behavior. Only the final fault class is eligible for model attribution after evaluator admission and blinded review.

The contribution is the chain, not any one component in isolation: **world changed → materially applied? → agent saw it? → authority valid? → agent acted correctly?** A failed link localizes an environment, disclosure, authority, or agent-fault candidate; it does not permit every red criterion to be counted as model behavior.

2. Research questions

- **RQ1 — Exact completion.** Does an identity-pinned frontier agent produce a perfect-rubric completion in each observed evolving workflow?
- **RQ2 — Two-attempt repeatability.** In exact provider × task cells with two same-runtime gradable attempts, does the model pass at least once, and does it pass both times? This is exploratory repeatability evidence, not a tail-reliability or general capability estimate.
- **RQ3 — Behavioral differentiation.** Which evidence, authority, temporal, and handoff obligations produce distinct, repeatable failure signatures?
- **RQ4 — Measurement validity.** Do deterministic checks, calibrated model-based interpretation, and blinded humans agree on which failures are attributable to the agent rather than the environment or evaluator?
- **RQ5 — Operational cost.** What calls, tokens, tool actions, wall time, and provider spend are required for a valid attempt, and how much missingness is caused by infrastructure rather than behavior?
- **RQ6 — Assisted-attribution stability.** When four independently pinned model judges inspect the same proof-bound evidence, how stable are their labels, cited anchors, and review priorities? This post-hoc question measures machine-review stability and triage value, not human validity or reasoning faithfulness.

2.1 Contributions and evidence status

This manuscript contributes four connected artifacts, with deliberately different claim strength:

1. **A proof-bound Universe composition.** Frozen task, model, runtime, evaluator, material-world roots, per-agent projections, action ledger, event/restore lineage, and disposition form one recomputable episode identity. The composition is implemented and mutation-tested; its superiority over alternative evidence systems is still a hypothesis.
2. **Four-way measurement fault localization.** The evidence chain separates world application, agent disclosure, authority, and downstream behavior before scoring. Scripted controls establish sensitivity; comparative incremental detection and human validity remain unmeasured.
3. **A fail-closed empirical protocol.** Model/runtime latching, evaluator admission, infrastructure censors, cohort-aware denominators, a public redacted evidence index, and non-reward diagnostic vectors make overclaim refusal executable rather than editorial.

4. **A live synthetic mortgage testbed.** Fifty-five physical attempts demonstrate the instrument under incomplete documents, retraction, authority conflict, simulated time, and handoff. The run is evidence about this edition—not yet a balanced model comparison, domain-valid benchmark, or novelty result.

3. Related work and measurement boundary

Dynamic and stateful agent benchmarks already establish that useful evaluation must go beyond static question answering. Gaia2/Agents Research Environments studies asynchronous changing worlds; InterruptBench studies user revisions and retractions; and tau-bench, tau2-bench, AppWorld, ToolSandbox, and OSWorld evaluate agents through actions that alter persistent environments [1–7]. Those systems occupy the broad territory of dynamic environments, interruptions, and stateful tool use. This work does not claim those ideas as new.

The compact comparison below makes the nearest-neighbor distinction explicit. It compares each cited paper's primary evaluation object rather than claiming that an unlisted implementation feature is absent.

System	Primary object in the cited work	Relationship to this paper
τ^2-bench [4]	Shared dual-control agent–user task	Binds application → projection → authority → judgment
AppWorld [5]	Stateful multi-app coding; state and collateral-effect tests	Adds time-indexed world/view roots, restore lineage, and typed refusal
ToolSandbox [6]	Stateful tool use; user simulation; milestone grading	Separates environment, disclosure, authority, and behavior faults
Gaia2 [1]	Asynchronous evolving worlds; write-action verification	Jointly recomputes occurrence, roots, projection, restore, evaluator, and disposition
BranchBench [16]	Branchable-database performance under branch–mutate–evaluate	Uses branches as evaluation interventions with whole-world comparability
Proof of Execution [8]	Authorized, tamper-evident, replayable execution	Adds semantic attribution: what changed, was seen, was authorized, and was correct
Conditionally Approved	**Attribution in changing-world agent evaluation**	**Tests the full application → visibility → authority → behavior chain with fail-closed denominators**

The narrower measurement problem is whether a benchmark can prove the chain from a declared occurrence to the material world transition, the exact projection visible to the agent, the eligible action boundary, any restore lineage, and the final judgment. Proof of Execution, CAVA, agent-native telemetry, W3C PROV, and recent runtime-evidence work establish adjacent execution, provenance, canonical-action, and telemetry foundations [8–14]. Agentic Data Environments and BranchBench make the complementary systems case that consequential agents operate over a closure of files, databases, processes, applications, APIs, memory, and derived state, while Data Flow Control studies semantic source-to-sink restrictions [15–17]. The method paper therefore tests a composition and its incremental detection value; it does not present hashing, logging, branching, provenance, or checkpointing as novel in isolation.

Reference [10] is a verified August 2026 preprint whose scope is agent-native, verifiable state-delta telemetry. We use it only as adjacent evidence architecture; we do not treat its existence as validation of this benchmark, claim its results as ours, or claim independent reproduction.

Benchmark-quality work also warns that a clean headline score can conceal broken tasks, evaluator leakage, contamination, or a saturated ceiling [18–20]. Conditionally Approved responds with mutation-tested evaluators, immutable preregistration, explicit infrastructure censors, runtime-stratified denominators, and human adjudication. These controls make the result more auditable; they do not substitute for domain-expert validity or a sufficiently powered behavioral study.

4. Benchmark design

4.1 Frozen five-task panel

Task edition	Core stressor	Required capability	Registered treatment
Document truth under pushback	Incomplete and conflicting document evidence	Reconcile exact pages, versions, claims, and current authority	Alder
Epistemic residue	Evidence observed, retracted, and replaced	Stop using dead evidence and re-ground the decision	Cedar
Authority and fair judgment	Two legitimate principals conflict	Apply authority scope, protect fairness constraints, and escalate correctly	Dogwood
Temporal portfolio control	Coupled files evolve around a commitment boundary	Coordinate scarce capacity, deadlines, and post-change rechecks	Fir
Honest handoff and verified commit	Work crosses an agent/session boundary	Preserve active constraints, uncertainty, provenance, and successor usability	Birch

The stories, people, organizations, values, documents, and policies are synthetic. They are not customer records, mortgage advice, production policy, or evidence of external-domain validity. The registered labels are display names; the public evidence index preserves every exact task and cell identifier.

4.2 Long-horizon execution

The safe scripted controls require 113–150 actions under full load. That observation establishes instrument solvability and the existence of a long execution path; it does not establish frontier difficulty. Model attempts may use fewer or more actions. Act count, verbosity, latency, and cost are diagnostic variables rather than reward criteria unless a task explicitly binds a deadline or resource limit.

4.3 Evolving-world evidence

Each declared occurrence binds its frozen event identity, exact eligible action boundary, before/after material roots, agent-visible projection, authority, restore generation, and previous occurrence-chain head. This lets the verifier distinguish environment application, disclosure, authority, restore, evaluator, and agent faults before an episode enters a model-performance denominator.

4.4 Formal episode and disposition contract

An episode is represented as

$$E_i = (T_i, M_i, S_i, W_{i,0}, A_i, \Omega_i, R_i, J_i),$$

where T_i is the frozen task edition, M_i the requested and resolved model identity, S_i the scaffold and runtime identity, $W_{i,0}$ the initial material world root, A_i the ordered act ledger, Ω_i the occurrence-witness chain, R_i the restore lineage, and J_i the evaluator edition and criterion vector. Every occurrence $\omega_{i,k}$ binds the declared event, eligible boundary, before/after roots, visible projection, authority, restore generation, and the previous occurrence digest. A verifier recomputes these bindings from released bytes; it does not trust a reported scalar.

Before scoring, every physical attempt receives exactly one disposition:

1. **gradable** — protocol, identity, runtime, world, and evaluator evidence close;
2. **infrastructure censor** — dispatch, transport, provider, or host evidence makes behavior unscorable;
3. **measurement defect** — the environment or evaluator cannot support the intended attribution; or
4. **pending adjudication** — deterministic observations exist, but their cause has not yet been assigned.

Only the first disposition may enter a behavioral denominator. The others remain visible, retain cost and receipt evidence, and cannot be silently retried into a preferred outcome.

4.5 Gradia Universes: methodological differentiators

Gradia Universes is the execution and evidence apparatus used to build and verify this benchmark; it is not the benchmark's name. Its purpose is to turn an episode from a transcript plus score into a versioned scientific object. Seven design choices matter for this study:

1. **Material world and visible world are separate commitments.** The runtime binds both the root-owned state transition and the exact projection available to the agent. An event cannot count merely because it appears in an operator log.
2. **Evolution is witnessed at an eligible act boundary.** Every occurrence binds what changed, when the change became visible, the before/after roots, authority, previous chain head, and restore generation. This localizes disclosure and stale- context defects that a terminal-state judge cannot see.
3. **Restore and replay preserve lineage rather than erasing history.** A restored world must name its parent, generation, and occurrence history. Counterfactual work can therefore distinguish a legitimate fork from an unrelated rerun.
4. **Identity is part of the measurement.** Requested and resolved model, task, evaluator, runtime, scaffold, Git commit, and sampling contract are frozen into the episode. Provider substitution or runtime drift is a measurement event, not an invisible implementation detail.
5. **Disposition precedes scoring.** Infrastructure censorship and measurement defects are retained with receipts but excluded from model denominators. This prevents transport failures from becoming apparent model failures and prevents retries from silently replacing inconvenient outcomes.
6. **Evaluators must survive mutations.** Positive controls establish a solvable floor; isolated one-defect probes test whether each criterion catches exactly its intended defect; current-source and post-change reread requirements prevent a plausible narrative from passing without observable evidence.
7. **Analysis is claim-gated.** Binary reward, criterion diagnostics, trajectory shape, human adjudication, causal inference, and training labels are separate evidence products. Analytics+ may summarize only the claims licensed by the exact evidence edition rather than upgrading correlations into mechanisms.

Dynamic worlds, interruption handling, snapshots, event logs, and provenance are all established prior art. The candidate research contribution is narrower: the composition above may provide incremental defect detection and more defensible behavioral attribution than terminal state, ordinary logs, or an unversioned judge alone. This paper demonstrates the instrument, its live execution inventory, and its refusal behavior. A comparative detection study and blinded human agreement are still required before that incremental-value hypothesis—or any novelty claim—is accepted.

4.6 Universe execution and proof pipeline

A benchmark task is only one frozen measurement inside a Universe. The Universe also binds the material state that can change, the actor-specific view of that state, executable tools, authority, events, logical time, evaluator identity, rights, and release posture. The distinction matters because the same task text can measure different behavior if any of those surrounding objects drift.

The execution pipeline has five ordered stages:

1. **Freeze:** approve the construct and bind task, material world, actor projection, tools and authority.
2. **Latch:** bind requested/resolved model, scaffold, runtime, sampling and code before dispatch.
3. **Execute:** append acts while root-authored events evolve material state.
4. **Witness:** bind each event to its roots, visible projection, boundary and restore lineage; finish with a terminal state and evidence-cited account.
5. **Disposition:** reverify the admitted evaluator and classify the physical attempt before any claim-gated analysis.

Five invariants govern the path:

1. **Identity closure.** Task, world, scenario, evaluator, model, scaffold, runtime, sampling and code identities are immutable before the relevant act.
2. **World closure.** Every scored side effect must be represented in the declared material world or named as uncovered; a terminal value cannot conceal an unsupported external mutation.
3. **Projection closure.** Evidence used to judge an agent claim must have been in that actor's exact visible projection at an eligible act boundary.
4. **Lineage closure.** Snapshot, restore and fork operations preserve the parent, generation, occurrence prefix and before/after roots.
5. **Disposition closure.** Infrastructure, measurement and pending-human states are decided before behavioral aggregation. A retry is a new physical attempt, never an overwrite.

These invariants turn "the agent was interrupted" into a set of falsifiable statements: the event was frozen, it was applied exactly once, material state changed as declared, the agent could observe the permitted projection, the action occurred after that projection, and the evaluator read the same edition.

One event, one declared action boundary, one auditable evolution chain

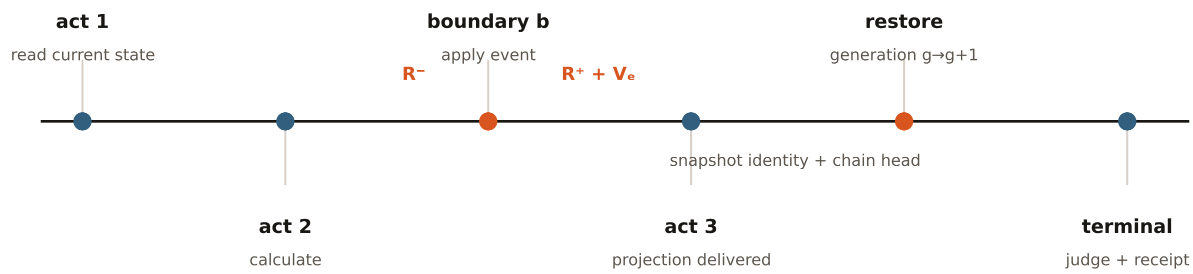

Figure 2: One witnessed episode binds logical boundaries, before/after roots, the actor-visible projection, restore generation, and terminal judgment.

4.7 Evolution-witness formalism

Let W_k be the material world before occurrence k, A_b the eligible act boundary, $P_{a,k}$ the projection visible to actor a, G_k the restore generation, and H_{k-1} the previous occurrence-chain head. An evolution witness is the self-bound object

$$\omega_k = \mathcal{H}(e_k, A_b, \mathcal{H}(W_k), \mathcal{H}(W_{k+1}), \mathcal{H}(P_{a,k}), q_k, G_k, H_{k-1}),$$

where e_k is the frozen event and q_k is its authenticated authority and visibility rule. Admission requires all of the following:

$$\begin{aligned}
&\text{eligible}(e_k, A_b) = 1,\\
&\text{apply}(W_k, e_k) = W_{k+1},\\
&\text{project}_a(W_{k+1}, q_k) = P_{a,k},\\
&\text{previous}(\omega_k) = H_{k-1},\\
&\text{generation}(\omega_k) = G_k.
\end{aligned}$$

This does not prove that the agent attended to or believed the projection. It proves the narrower prerequisite that the declared world change occurred and identifies the exact information the agent could have used. That separation is what permits a later behavioral claim to be contradicted, supported, unobservable, or indeterminate without guessing private cognition.

In a multi-actor Universe, each $P_{a,k}$ is an actor-specific causal view rather than a copy of the root log. Hidden events are absent, not merely redacted after delivery. A future noninterference study can therefore ask whether an actor's action changes when only facts outside its admitted causal past change. The present panel implements single-agent projection evidence; it does not report that future result.

4.8 Branch comparability and counterfactual discipline

A branch is scientifically useful only if siblings descend from one admitted parent. Let C be a closure manifest over material tables, documents, scenario state, queues, clocks, actor memory and every other declared mutable component. Two child worlds W^0 and W^1 are an eligible pair only when:

$$\text{parent}(W^0) = \text{parent}(W^1), \quad \text{closure}(W^0) = \text{closure}(W^1) = C,$$

and their intervention manifest declares one changed factor while all protected identities remain equal. The evidence must bind the branch operation, child roots, actor projections, runtime identity, evaluator, and terminal utility. Missing process, sidecar, queue, clock, cache, credential or external replay state is not silently assumed equal; it is a named closure blocker.

This contract supports three different questions that must not be collapsed:

- **Descriptive sensitivity:** did behavior differ between witnessed siblings?
- **Recoverability:** could an admitted reference continuation recover from a recorded point?
- **Causal credit:** did substituting one act, while the policy rejoined under a valid paired design, change terminal utility?

Only the first can be reported from an ordinary paired fork. Recoverability needs a pinned reference policy; act-level blame or training-grade advantage needs repeated isolated substitutions, stable deltas, parity, human/evaluator calibration, independent review, holdout and appropriate training rights. This paper reports no causal credit or training label.

4.9 What is established, composed, and still only hypothesized

The research claim is intentionally componentwise:

- **Stateful agent environments are established.** This composition freezes world, task, runtime and evaluator identity together. That binding is implemented; superiority over existing environments is untested.
- **Asynchronous change, interruption and retraction are established.** The added measurement object binds occurrence, material application, actor projection and act boundary. Scripted sensitivity passes; incremental detection is untested.
- **Logs, telemetry and provenance are established.** The composition keeps the actor-visible chain distinct from the root/auditor chain. Its engineering evidence is recomputable; comparative value is not yet measured.
- **Snapshot, restore and branching substrates are established.** Gradia admits a branch only with parent, generation, occurrence and declared-closure lineage. Local controls pass; provider-wide whole-world closure remains unproved.
- **Rubric evaluation, judge calibration and mutation testing are established.** This evaluator adds isolated one-defect probes plus current-source and post-change reread gates. Deterministic probes pass; human agreement is pending.
- **Counterfactual oversight has extensive prior work.** Concealed witnessed forks are a candidate way to test account consistency. This is a research program only; the present paper makes no truthfulness or deception claim.
- **Long-horizon tasks and empirical pass rates are established.** This protocol separates difficulty from solvability, engagement and infrastructure missingness. The current panel is incomplete, preliminary machine evidence and supports no calibrated tier.

The candidate contribution is therefore not "we invented dynamic environments" or "hashes make agents honest." It is a testable evidence composition: bind world evolution, actor visibility, action boundaries, restore/branch lineage and admitted judgment tightly enough to detect defects that terminal state and ordinary logs miss. The falsifier is straightforward. If a strong baseline catches the same planted defects with comparable false-positive rate, overhead and reconstructability, or if blinded humans cannot reliably validate the additional distinctions, the incremental-value claim fails. That comparative study remains the central next methodological experiment.

5. Evaluation contract

5.1 Official reward

Official reward is 1 only when every registered binary requirement passes. It is 0 otherwise. Criterion vectors, branch probes, tool-use traces, and narrative analyses are diagnostic evidence—not hidden partial credit. A correct and complete result is not penalized merely for taking more actions.

Formally, if $g_{i,1}, \dots, g_{i,K_i} \in \{0,1\}$ are the registered applicable gates, the official episode outcome is

$$y_i = \prod_{k=1}^{K_i} g_{i,k}.$$

The conjunction prevents a strong final narrative from compensating for stale evidence, missing authority, an unauthorized irreversible act, or an unusable handoff. Separately reported criterion coverage can help diagnose where an episode failed, but it is not partial credit and cannot change y_i.

5.2 Evaluator admission

The evaluator package must demonstrate:

1. a positive control for every task edition;
2. isolated one-defect mutations for each criterion;
3. current-source citations backed by root-owned read evidence;
4. post-boundary re-reading of every materially changed resource;
5. exact task, runtime, world-root, and evaluator-edition binding;
6. blinded independent human review of the criteria and representative outcomes; and
7. a fail-closed disposition for unresolved measurement defects.

Deterministic red criteria are initially **unresolved observations**. They become model-attributable failure modes only after adjudication rules exclude environment, provider, and evaluator defects.

5.3 Panel, amendment, and denominator

The original plan crossed four identity-pinned provider/model cells with five task editions and five gradable attempts per cell. It was not completed. The cost-cap amendment instead defines 20 exact provider × task cells, targets two gradable attempts per cell, and limits the entire execution to 55 physical attempts. The amendment froze at 07:42:08 UTC on 24 August 2026 **after** some attempt artifacts and their zero outcomes existed and had been inspected. It is therefore a post-observation operational cost stop, not an untouched prospective amendment. Within each cell, however, selection was mechanical and outcome-blind: choose the lowest two gradable ordinals. Because every gradable outcome was zero, stopping could not have selected observed successes. The panel remains exploratory because the timing, missingness, and runtime split still matter. Infrastructure failures retain their receipts and costs but do not enter behavioral denominators. Within a cell, the lowest two gradable ordinals are selected; later gradable attempts remain disclosed as overage and cannot replace an earlier result. An exact-cell pair is reportable only if both selected attempts bind the same task, evaluator, model identity, Git commit, and runtime-implementation digest.

The execution wrapper was allowed to remain active while the repository advanced. The post-run verifier therefore partitions attempts by exact runtime cohort and refuses pooled panel claims if more than one cohort is present. A later code gate latches the panel commit before dispatch and rejects any child that drifts. The historical cohort split is retained as a measurement defect, not edited away. No replacement attempt will be appended to this exhausted edition. A follow-on edition must first pass provider authentication/protocol preflight, latch the exact manifest, model, evaluator, runtime, and commit before dispatch, freeze one runtime cohort, and preregister replacement and stopping rules before outcomes are observed.

6. Locked analysis plan

For each provider × task cell, report:

- physical attempts, gradable attempts, and infrastructure exclusions;
- the exact runtime cohort of every selected attempt;
- `pass@2`: whether at least one of the two selected same-runtime attempts succeeds, only where an eligible pair exists;
- `pass^2`: whether both selected same-runtime attempts succeed, only where an eligible pair exists;
- single-attempt outcomes where exactly one gradable attempt exists, without relabeling them as pass@2;
- infrastructure exclusions and reasons;
- calls, input/output tokens, tool actions, wall time, and cost;
- registered criterion-failure frequencies, initially labeled unresolved; and
- identity-blinded Analytics+ interpretations with cited trajectory evidence and human disposition.

Across the panel, report inventory summaries without pooling away task, provider, or runtime identity. A balanced statistic exists only if all 20 exact cells have two eligible attempts under one common runtime cohort.

No significance test, causal claim, provider ranking, pass@5 claim, difficulty claim, or novelty claim may be added after seeing outcomes unless it belongs to a separately preregistered study.

For an eligible exact cell c with the two selected binary outcomes $(y_{c,1}, y_{c,2})$, this draft uses two deliberately different descriptive quantities:

$$\text{pass@2}_c = \max(y_{c,1}, y_{c,2}), \qquad \text{pass}^2{}_c = y_{c,1} y_{c,2}.$$

The first asks whether either observed attempt passed; the second asks whether both did. With only two attempts, neither is a stable tail-reliability or population capability estimate. No value is emitted if the two attempts differ in frozen task, model, evaluator, Git commit, or runtime-implementation digest.

7. Results

Preliminary machine results. This section is generated from verification artifact `b8e013fc05 b5f9868a892f89a2a0986d67978c68153b47d9b1756fbbec3b5777`. It reports exact execution and scoring facts, not human-attributed failure modes.

7.1 Completion and exclusions

The cost-capped execution completed at 14:24:51 UTC on 24 August 2026 after reaching its amended hard ceiling of **55 physical attempts**. Of those, **37** produced a complete, gradable terminal artifact and **18** were infrastructure exclusions. The machine-scored perfect-rubric outcome was **0/37** among gradable attempts. This is an exact observation about the registered artifacts—not an estimate of broad model capability and not yet an attribution to model behavior.

The 18 exclusions remain in the physical-attempt and cost ledgers but never enter behavioral denominators. Their terminal causes were seven OpenAI HTTP 401 authentication refusals, two Anthropic HTTP 529 overload failures after bounded retries, six read timeouts, one OpenAI read error, and two xAI responses that completed transport but did not yield an admitted text payload. Excluded attempts accounted for **$173.35** of recorded spend. Seven OpenAI authentication exclusions failed before a billable response and recorded zero spend; several other exclusions occurred only after long partial trajectories.

Model	Physical	Gradable	Infra	Perfect passes	Spend
Opus 5	14	8	6	0	$281.50
Gemini 3.1 Pro	13	13	0	0	$2.99
GPT-5.6-sol	18	10	8	0	$308.62
Grok 4.6	10	6	4	0	$112.77
Total	**55**	**37**	**18**	**0**	**$705.88**

The provider rows are execution inventory, not a reliability or capability ranking. The providers saw different cell mixes, the panel crossed runtime cohorts, and the sample is small and outcome-incomplete.

7.2 Exact-cell outcomes and repeatability

The verifier reconstructed 20 registered provider-by-task cells. Six cells contained two eligible gradable attempts under one exact runtime cohort. All six emitted `pass@2 = false` and `pass^2 = false`: neither selected attempt passed. Nine cells contained one gradable attempt, and each of those observed outcomes was also zero. Five cells had no gradable attempt. Sixteen additional gradable executions were retained as disclosed overage rather than reassigned to missing cells.

The six complete cells provide a legitimate complete-case descriptive subset:

Model	Task	Outcomes	pass@2	pass^2
Opus 5	Document truth	`[0, 0]`	false	false
Gemini 3.1 Pro	Document truth	`[0†, 0†]`	false	false
GPT-5.6-sol	Document truth	`[0, 0]`	false	false
Grok 4.6	Document truth	`[0, 0†]`	false	false
Gemini 3.1 Pro	Epistemic residue	`[0†, 0†]`	false	false

Model	Task	Outcomes	pass@2	pass^2
GPT-5.6-sol	Epistemic residue	`[0, 0]`	false	false

† marks a deterministic **low-engagement protocol trace**: tool actions were below ten percent of the scripted reference control and the episode observed no source, attachment, or world event. The dagger is a post-hoc disposition annotation, not a reward adjustment or exclusion. Every daggered attempt remains an official gradable zero and remains in the same denominator; its behavioral interpretation is simply not presented as comparable long-horizon engagement before human review.

All six pairs share registered execution commit `9084b884` and runtime implementation digest prefix `7449c698`. Appendix A preserves the exact identities; the shortened display labels above are only for readability.

The **Document Truth Under Pushback** slice is the cleanest cross-provider result: all four providers supplied two gradable attempts on the same task, evaluator, Git commit, and runtime implementation. None of the eight selected attempts passed, so all four exact provider cells have `pass@2 = false` and `pass^2 = false`. This is a complete same-runtime result for one synthetic task; it is not a ranking and cannot be extrapolated to the other four tasks. The two Epistemic Residue pairs add four same-runtime gradable attempts for Gemini and OpenAI, also with no perfect-rubric completion.

Cell coverage state	Cells	Permitted statement
Exact same-runtime pair	6	Exact observed `pass@2` and `pass^2`; both false in all six
One gradable attempt	9	One exact binary outcome; all nine were zero
No gradable attempt	5	Missing; no behavioral statistic

The result does not support a pooled `0/20` cell score. It supports 37 exact machine-scored failures, six exact paired observations, and a refusal to synthesize the missing denominator. Three runtime cohorts were present: commits `50a62f9e`, `9084b884`, and `b4966b1c`, spanning two runtime implementation digests. The panel therefore fails its common-runtime requirement.

7.3 Cost and trajectory shape

Across all 55 physical attempts, the ledger recorded **3,617 provider calls**, **99,382,990 input tokens**, **6,080,548 output tokens**, and **$705.88** in provider spend. Those totals include infrastructure exclusions because their calls, tokens, and costs physically occurred. The **2,609 accepted or refused tool actions** and **5,325 transcript turns** come only from the 37 complete gradable episode records. Infrastructure artifacts retain 997 partial provider-turn records, but those partial records are not promoted into official episode transcripts or tool counts.

The six complete pairs expose substantial trajectory heterogeneity behind the identical terminal outcome:

Model / task	Calls	Tools	Turns	Input tokens	Output tokens	Spend
Opus 5 / Document	152	150	308	3,849,883	297,486	$26.69
Gemini 3.1 / Document	5	4	8	34,110	3,337	$0.20
GPT-5.6 / Document	201	200	409	4,591,400	64,078	$38.65
Grok 4.6 / Document	147	145	295	5,003,222	634,642	$27.63
Gemini 3.1 / Residue	6	4	8	40,228	5,277	$0.26
GPT-5.6 / Residue	305	303	618	8,328,831	103,381	$69.73

For Document Truth, the selected Gemini pair used four tool actions while the OpenAI, Anthropic, and xAI pairs used 200, 150, and 145 respectively. Both Gemini attempts and the second selected xAI attempt met the low-engagement descriptor. The Epistemic Residue pairs show the same qualitative contrast: four versus 303 tool actions, and both selected Gemini attempts were low-engagement. Across the complete panel, 13/37 gradable attempts and 7/21 selected primary attempts met that rule. A short trace can represent protocol non-engagement rather than efficient reasoning; therefore these rows must not support a provider capability comparison. The result instead shows that a terminal scalar collapses materially different execution shapes and that the action/evidence ledger exposes a reviewable distinction.

7.4 Finding: a repeatable branch-surface short-circuit

The Gemini records make the mechanism inspectable rather than speculative. In all five Document Truth attempts and all five Epistemic Residue attempts, the policy queried `branch.inspect` and then attempted a terminal submission after only two tool actions. It read no portfolio source, attachment, or applied world event. The prompt had explicitly required the complete six-case base portfolio before the branch account, so the behavior is a repeatable short-circuit through the branch surface—not evidence of completing the underlying job efficiently. The generic student scaffold's `submit` action was routed to the base terminal interface while the model supplied branch-account fields, producing an accepted-for-grading but incomplete world and the frozen `INVALID` outcome. This can be read as model protocol failure, scaffold affordance ambiguity, or their interaction; the current edition cannot identify which. Repeating the same bytes would add little. A fair replacement study must create a new edition that (i) makes the base-versus-branch terminal distinction explicit, (ii) refuses branch submission until the base terminal contract is complete, and (iii) preregisters the comparison before any new outcome is observed. The frozen attempts remain unchanged.

Subsequent engineering remediation—not a live-model result. A digest-distinct private `v4r1` edition now implements those three requirements at Gradia commit `c5e0f72`. The base terminal accepts only the exact root-owned `draft_sha256`; the branch terminal refuses until that base receipt exists; and generic submit routes to the branch terminal. Its merged-tree control suite passed 10/10 tests: one exact historical-shortcut reproduction, two malformed-or-premature terminal refusals, five complete scripted-world positives, one provider/model identity and manifest drift check, and one key-free write-once preflight check. The five positive scripts contain 138, 150, 153, 153, and 117 acts respectively, and act count is not a reward input. This establishes solvability of the corrected protocol and sensitivity to the observed interface defect. It does **not** establish that Gemini, or any live model, would engage successfully with `v4r1`; no paid replacement attempt exists. The private correction is named for auditability but is not public-repository reproduction evidence. A fresh result still requires independent edition review, an exact-model smoke receipt, a prospectively latched manifest, and a separately labeled cohort.

The scale matters operationally but is not itself a difficulty metric. A long trace can reveal expensive context accumulation, repeated reconciliation, or provider fragility; only an admitted outcome can establish exact task completion, and only a balanced reviewed panel can support a broader behavioral comparison.

7.5 Criterion-level observations

The 37 gradable episodes carried **1,375 applicable machine-diagnostic criterion assignments**. Of these, 486 were green and 889 were red. This 35.3% diagnostic coverage is useful for locating evidence, but it is **not a 35.3% score**: official reward remains the binary conjunction, and every episode received 0. A green criterion also does not prove that adjacent obligations or the terminal work product were correct.

Task	Gradable attempts	Machine green	Machine red	Applicable assignments
Document truth under pushback	18	232	434	666
Epistemic residue	12	188	280	468
Authority and fair judgment	3	34	77	111
Temporal portfolio control	3	29	64	93
Honest handoff	1	3	34	37
Total	**37**	**486**	**889**	**1,375**

The red assignments cluster on concrete observable surfaces:

Diagnostic family	Red assignments
Document truth	163
Temporal control	159
Derived-state freshness	119
Belief revision	85
Authority	82
Evidence provenance	81

Diagnostic family	Red assignments
Terminal integrity	77
Portfolio reasoning	66
Action safety	34
Completion	13
Handoff continuity	7
Fairness safety	3

The most frequent exact red criteria were missing or late deadline control (37/37), unresolved rate-lock risk (37/37), stale authoritative market state after repricing (35/37), unsafe use of the document-source outage interval (35/37), unevidenced handling of unverified pressure (34/37), uninspected material documents (34/37), terminal packet mismatch (34/37), and unreconciled post-change histories (34/37). These denominators are all gradable attempts because those criteria are shared across the composed v4 contract; task-local criteria have smaller applicable denominators and must be read from the released index.

These are **machine-observed failure surfaces**, not yet model-attributed failure modes. A short non-engaging trace, an evaluator defect, or an environment disclosure defect can generate several red criteria at once. Each red assignment must therefore be labeled independently as `model_failure`, `measurement_defect`, or `unresolved`, with evidence citation and rationale. A measurement defect cannot be repaired by relabeling the historical attempt; it creates a new evaluator or task edition and a separately identified replacement run.

7.6 Analytics+ findings

Analytics+ can already localize task, criterion, phase, source, authority, event, engagement, and provider-failure surfaces. The current pre-adjudication register is:

Observed surface	Evidence	Permitted interpretation now
Infrastructure censor	18/55 physical attempts	Operational reliability and cost evidence; never model failure
Low engagement	13/37 gradable, including 7/21 primary	Protocol-engagement annotation; not efficiency or incapability
Gemini branch short-circuit	10 attempts across two tasks followed the same two-action shape	Repeatable model/scaffold interaction; causal share unresolved
Near-universal red criteria	Six exact criteria red on 34–37/37 attempts	Highest-priority measurement audit; not mass model attribution
Substantive but non-passing traces	Complete cells range from 145 to 303 tool actions per pair while all strict outcomes are zero	Genuine terminal non-completion with mechanism pending human review
Runtime drift and missing cells	Six exact pairs, nine singletons, five unobserved cells	Cohort comparability defect; no pooled provider ranking

These entries are failure **surfaces**, not yet human-validated failure modes. Temporal and document-truth obligations account for 322/889 red assignments, and identical terminal zeros hide radically different interaction paths. Analytics+ is not allowed to infer a behavioral mechanism from those facts. Evidence-cited model-failure interpretations will be generated only after reviewer dispositions are sealed, and every interpretation must preserve contrary evidence, missingness, runtime cohort, engagement posture, and denominator eligibility.

7.7 Four-judge attribution and evidence-alignment stress test

The strict machine evaluator identifies red criterion surfaces; it does not assign their cause. To test whether one LLM judge could safely bridge that gap, four independently pinned judges—Claude Opus 5, Gemini 3.1 Pro Preview, Grok 4.6, and GPT-5.6-sol—received all 889 blinded assignments with the same frozen transcript, tool, event, root, and criterion evidence. Each judge returned one of `model_failure`, `measurement_defect`, or `unresolved`, plus confidence, a concise audit rationale, an uncertainty or alternative, a reviewer focus, and one or more evidence citations. Raw chain-of-thought was neither requested nor retained. Every vote was advisory, and the alignment verifier recorded **zero score mutations**.

The stability result is stark. Only **232/889 (26.1%)** assignments were unanimous; 384 had a 3–1 majority, 185 had a 2–1–1 plurality, and 88 tied 2–2. Descriptive Fleiss' κ was **0.151**. The panel routed 126 assignments to critical and 556 to high human-review priority, so **682/889 (76.7%)** remained priority review work. A unique model-failure plurality existed for 594 assignments, while 121 favored a measurement defect, 86 favored unresolved, and 88 had no unique disposition. These are machine votes, not validated failure counts.

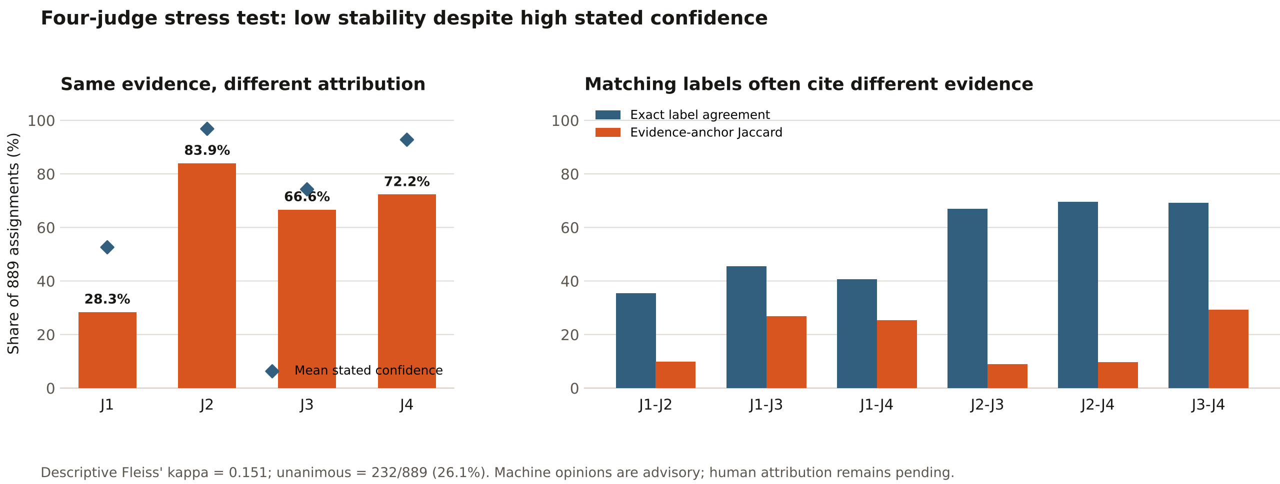

Figure 3: Four judges received the same evidence but produced materially different attribution rates; matching labels frequently cited different evidence anchors.

Judge	Model failure	Measurement defect	Unresolved	Mean confidence
Claude Opus 5	252	209	428	0.526
Gemini 3.1 Pro Preview	746	107	36	0.968
Grok 4.6	592	128	169	0.742
GPT-5.6-sol	642	178	69	0.928

The share labeled model failure ranged from 28.3% to 83.9%, even though all four judges inspected identical bytes. Self-reported confidence did not resolve this: the most model-failure-prone judge also reported 0.968 mean confidence. The result does not identify which judge was right. It shows that a single judge's confidence cannot substitute for independent attribution.

Judge pair	Exact label agreement	Mean cited-anchor Jaccard	Label agreements with zero shared anchor
Claude–Gemini	35.4%	0.098	77.8%
Claude–Grok	45.4%	0.268	61.6%
Claude–GPT	40.5%	0.252	60.3%
Gemini–Grok	66.8%	0.089	86.4%
Gemini–GPT	69.5%	0.095	85.1%
Grok–GPT	69.1%	0.292	61.4%

Label agreement was therefore much more common than shared-evidence agreement. Depending on pair, **60.3%–86.4%** of exact-label agreements shared no exact evidence-anchor identifier. Distinct anchors can support the same valid conclusion, so zero overlap is not itself contradiction. It is nevertheless a warning that consensus labels can conceal different evidentiary paths. Proof-bound citations make that divergence inspectable; a scalar label cannot.

Criterion-family distribution The family distribution separates three questions that a raw failure count merges: how often a surface is red, whether judges agree, and whether a unique model-failure plurality exists. "Priority" is critical or high routing. "No unique MF" combines a measurement-defect plurality, unresolved plurality, or tie. None is partial credit.

Criterion family	Red	Share of 889	Unanimity	No unique MF	Priority
Document truth	163	18.3%	41.7%	16.6%	60.1%
Temporal control	159	17.9%	22.6%	25.2%	78.0%
Derived-state freshness	119	13.4%	11.8%	41.2%	88.2%
Belief revision	85	9.6%	21.2%	30.6%	81.2%
Authority	82	9.2%	23.2%	57.3%	90.2%
Evidence provenance	81	9.1%	11.1%	59.3%	88.9%
Terminal integrity	77	8.7%	27.3%	28.6%	72.7%
Portfolio reasoning	66	7.4%	18.2%	27.3%	81.8%
Action safety	34	3.8%	64.7%	50.0%	58.8%
Completion	13	1.5%	46.2%	7.7%	53.8%
Handoff continuity	7	0.8%	57.1%	0.0%	42.9%
Fairness safety	3	0.3%	100.0%	0.0%	0.0%

Document truth and temporal control contributed 322/889 red surfaces (36.2%), but volume and attribution uncertainty separated sharply. Document truth had 41.7% unanimity and only 16.6% without a unique model-failure plurality; evidence provenance had 11.1% unanimity and 59.3% without one. A frequency-only failure taxonomy would therefore prioritize common behavior while missing the families most likely to contain measurement ambiguity.

Task-normalized distribution Applicable criterion counts differ by task, so red counts alone are not comparable. The table reports diagnostic density and review posture while preserving the strict binary reward. The Handoff row contains one attempt and is not a stable estimate.

Task	Red/applicable	Diagnostic red rate	Red/attempt	Unanimity	Priority
Document truth	434/666	65.2%	24.1	27.4%	74.4%
Epistemic residue	280/468	59.8%	23.3	15.0%	87.9%
Authority	77/111	69.4%	25.7	29.9%	72.7%
Temporal control	64/93	68.8%	21.3	39.1%	68.8%
Honest handoff	34/37	91.9%	34.0	67.6%	38.2%

The most valuable review hotspots were instrument-facing rather than simply frequent. Runtime binding and tool-receipt-chain integrity had no unique model-failure plurality in 100% of 19 assignments each; root-event application did so in 93.1% of 29; and narrative-versus-authority separation in 100% of 19. Unanimity was 0%, 5.3%, 3.4%, and 31.6% respectively, and all four routed at least 94.7% of assignments to critical or high review. This is a concrete proof-bound use case: audit the runtime, receipt, event, and authority evidence before accusing the agent. The public JSON preserves their exact registered criterion identifiers.

At the task level, Epistemic Residue combined the lowest unanimity (15.0%) with the highest priority routing among multi-attempt tasks (87.9%). The single Honest Handoff attempt reached 67.6% unanimity, so its apparent stability cannot be generalized. These patterns are consistent with the hypothesis that changing evidence, visibility, and derived state create attribution questions a terminal evaluator cannot settle alone.

This analysis tests three bounded value hypotheses:

1. **Terminal-score insufficiency is supported descriptively.** Identical zeros concealed two orders of magnitude in tool use, 12 criterion families, and materially different attribution profiles.
2. **Single-judge sufficiency is contradicted for this packet.** Low unanimity, $\kappa = 0.151$, and large vote-profile spread show that one judge is not a stable causal authority, even when confident.
3. **Proof-bound triage feasibility is demonstrated.** Every vote cited frozen evidence, contrary evidence remained visible, 121 potential measurement defects surfaced, and scores stayed immutable. Incremental accuracy or reviewer-time savings versus transcript-only review remains unproven and requires a prospective expert-controlled ablation.

The machine panel is therefore an **accelerator, not an adjudicator**. It can rank review urgency, surface disagreements, and hand experts evidence-cited alternatives. It cannot turn plurality into truth or replace the separately sealed two-human study. The public alignment summary preserves the exact aggregates, model pins, pairwise metrics, and source digests without releasing private traces.

7.8 What the cost-capped panel established

The run established seven facts.

1. The five tasks did not produce an immediate frontier ceiling: 37 complete attempts yielded no perfect-rubric completion.
2. The workloads exercised genuinely long trajectories, including thousands of provider and tool interactions across the panel.
3. Infrastructure reliability was itself material: 18/55 physical attempts were not valid model outcomes.
4. Runtime drift and uneven cell coverage made an apparently simple aggregate scientifically ineligible; the verifier caught and refused it.
5. The remaining uncertainty is now concentrated in a human-reviewable question: which deterministic red criteria represent agent behavior, and which reveal measurement defects?
6. Identical terminal scores concealed two-order-of-magnitude differences in tool activity and transcript length across complete cells, motivating process-level analysis without turning trajectory length into reward.
7. A deterministic non-reward engagement descriptor separated 13 low-engagement gradable traces from substantive evidence interaction without rewriting their official outcomes or denominator eligibility.

The run did **not** establish that all four model families have zero capability on the underlying work, that one provider is more reliable or capable than another, that the tasks are externally valid mortgage work, or that the proposed research constructs are novel. Those conclusions require the next gates rather than stronger language around the present data.

8. Validity and limitations

Five synthetic tasks and a cost-capped, runtime-stratified panel cannot establish mortgage-domain validity, broad agent capability, tail reliability, training utility, or real-world safety. A wall of zero scores may identify a floor, an over-constrained evaluator, or both; it does not by itself establish that the tasks are valuable. A wall of perfect scores may identify a ceiling; it does not justify making tasks arbitrarily obscure. Follow-on editions should be admitted from observed failure evidence, expert review, and controlled anchor tiers rather than post-hoc difficulty chasing.

The registered treatment cells do not alone identify causal effects of retraction, phase, authority, or handoff. Those claims require exact seed-paired controls or counterfactual branches, repeated executions, closure evidence, and a separately admitted analysis.

The observed infrastructure exclusions are not missing completely at random. Longer episodes create more opportunities for transport failure, and providers encountered different failure types and task mixes. Excluding those attempts is necessary to avoid false model failures, but it can bias the observed gradable set. Replacement attempts must therefore be preregistered under one frozen runtime after provider preflight; they cannot be appended to the exhausted 55-attempt panel.

Gradable does not imply comparable engagement. The engagement descriptor is deterministic and evidence-bound, but it was defined after outcomes were available. It is therefore an interpretability annotation rather than a confirmatory covariate, reward, or exclusion rule. Its next-edition threshold and analysis must be frozen before dispatch, and low-engagement cases require blinded protocol-engagement review before any behavioral interpretation.

9. Ethics, rights, and release

The public benchmark assets are synthetic. Model outputs, raw trajectories, derived publication, redistribution, and training use are separate rights surfaces and must be reviewed per artifact and provider. Public release requires the exact signed edition, source and evaluator identities, redaction decision, independent review, and a machine-verifiable release receipt. No customer or partner attribution is permitted without written authorization tied to the exact released bytes.

10. Reproducibility

This public draft includes the frozen method, aggregate tables, exact plan and amendment bindings, verification digests, a rendered PDF, and a redacted 55-row evidence index. The index exposes per-attempt artifact identity, disposition, analysis cohort, runtime cohort, usage, cost, machine criterion vector, and non-reward engagement descriptor. A public reader can recompute every aggregate in Sections 7.1–7.5 with `gradia-universe cost-capped-verify`; the verifier refuses index, chain, count, diagnostic, and engagement drift. Raw provider responses and full trajectories remain private pending per-artifact rights, redaction, and release review, so semantic reinspection of quoted behavior remains a human-controlled private gate rather than public reproducibility. The code, synthetic fixtures, verification commands, reference receipts, preregistrations, and current manuscript are available at github.com/rudycelekli/gradia-universes-work-sample.

11. Integrated method validation

This paper is the complete research report: Universe method, engineering validation, live mortgage testbed, empirical inventory, limitations and research program are one claim-bounded artifact. The sections below distinguish what the public verifier already demonstrates from the comparative studies that remain open.

The scope boundary with the separately tracked *Interruptible Universes* manuscript is deliberate. This paper owns the benchmark contract, the integrated method needed to audit it, and the live cost-capped findings. *Interruptible Universes* owns the future Study A comparison of evolution-witness evidence against admitted baselines on mutant runtimes. Shared definitions are repeated only where this report must be self-contained; no result in one manuscript is evidence for the other's open study.

11.1 Trust boundary and adversary

The verifier trusts the released reference implementation, canonical encoder and the bytes it is asked to verify. It treats agent output, tool arguments, model rationales, reported roots, occurrence claims and aggregate prose as untrusted. It recomputes identities and refuses a mismatch rather than repairing it. Under that boundary, a valid witness supports the statement that the declared event, recorded state transition, visible projection, action boundary, restore lineage and judgment are mutually consistent with deterministic replay.

It does **not** prove that an external source was truthful, the model read or understood a notice, a host was uncompromised, a caller-declared identity was authenticated by hardware, or a hash was signed by a known principal. Those require separate source authentication, isolation, workload identity, external sensing and signature evidence. A party with authority to rewrite both the implementation and all unsigned evidence could construct a new internally consistent history; exact release authorization and remote anchoring address different threats.

11.2 Scripted floor, unsafe controls, and replay

Before paying for model behavior, the public harness runs three deterministic policies over five reference fixtures. `interrupt_safe` re-reads authoritative state after change; `stale_context` commits from cached state; `message_credulous` promotes recommendation-shaped content without sufficient authority. These are planted program behaviors, not simulated people or language models.

Scripted policy	Exact passes	Constructed episodes
`interrupt_safe`	5	5
`stale_context`	2	5
`message_credulous`	1	5

The 15 episodes contain 80 recorded acts, 12 event occurrences, three restore receipts and zero environment failures. Replay regenerates the panel digest `8fe207d2394c15f8db07e01d33f350997b1915aab147eb1c7ab199 2804b620ff`. The safe floor passing and planted unsafe controls failing establish local harness sensitivity and solvability. They are not samples from a population, a difficulty tier or an effect estimate.

Two controlled traces explain why process evidence matters. In a stale-policy trace, the root policy changes at a witnessed boundary, but the unsafe policy cites the old source and submits the pre-event root; recommendation, current-root, adaptation and evidence criteria localize the failure. In an authority-conflict trace, the final recommendation happens to be correct but relies on an unverified message; outcome correctness passes while authoritative-evidence fails. A terminal answer alone cannot make that distinction.

11.3 Mutation battery and evidence-policy baselines

The current engineering preflight reconstructs 26 isolated forks from five valid parents. Each fork records its parent digest, mutation family, changed JSON paths, unchanged-leaf manifest and expected representation. The public verifier compares five evidence projections:

1. terminal state only (`T`);
2. ordinary event log plus terminal (`L+T`);
3. selected action milestones plus terminal (`M+T`);
4. an engineering proof-of-execution-style causal projection (`P+T*`); and
5. the full evolution witness (`W`).

Projection	Forks	Planted change represented	Planted change absent	Faithful-parent change
`T`	26	0	26	0
`L+T`	26	12	14	0
`M+T`	26	7	19	0
`P+T*`	26	15	11	0
`W`	26	26	0	0

The report digest is `f891ac53e2212051fc8287194f9a380adb214f6ed52113e3e7d1f04ee80ceb34`. These counts measure whether the generator-authored changed field remains present in each representation—an information-availability ceiling. They are **not** blinded detector outcomes and do not prove incremental validity. `P+T*` is deliberately called "style": it is not claimed as a faithful reproduction of Proof of Execution.

The mutation families cover logged-but-unapplied events, altered visible content, altered authority, duplicate or lost delivery after restore, broken predecessor links, repaired terminal states that conceal invalid intermediate state, and wrong action boundaries. Expected-miss controls include contamination outside declared closure and an evaluator implementation that contradicts its own frozen criterion. Benign metamorphic controls include JSON reordering, equivalent timestamp rendering, causally valid reordering of independent occurrences and bijective entity renaming. A method that rejects those harmless re-projections is presentation-sensitive rather than semantically invariant.

The frontier evaluator adds five positive admissions and 44 isolated one-defect probes. The phase/authority construction layer adds ten exposed PRE-RESULTS candidates and 100 isolated construction probes. These tests show that exact planted defects are caught under the authored fixtures. They do not establish judge–human agreement, live behavioral effects or scientific novelty.

11.4 Confirmatory ablations and falsification plan

The primary future method study applies the same isolated invalid fork to every evidence policy and asks whether the policy rejects it and localizes the first faulty origin. A strong comparator is required; ordinary logs alone are intentionally not the straw-man baseline. Starting from `W`, the study removes the event digest, before root, after root, projection digest, action boundary, restore generation, predecessor link, act ledger and terminal judgment one at a time. A component is empirically justified only if its removal changes detection, localization, false rejection, replay behavior or cost on a preregistered mutation family.

The primary comparison is paired invalid-episode detection between `W` and the strongest admitted baseline, with exact-origin localization and faithful-parent false rejection as essential counterweights. Evidence bytes, verifier time and reviewer time must be reported so a larger receipt cannot win merely by retaining everything. Multiple mutations from one parent remain clustered rather than being misreported as independent tasks.

The method claim fails or narrows if a strong baseline catches the same defects with comparable false-positive rate, localization, cost and reconstructability; if valid runtime adapters cannot preserve the required semantics; or if blinded humans cannot reliably validate the distinctions. A null comparison is useful: it would identify which witness fields are redundant and where simpler state or execution evidence is sufficient.

11.5 Claim ladder

The present evidence licenses only the lower rungs:

1. **Implemented:** canonical objects, replay, mutation probes and the public cost-capped index execute as described.
2. **Engineering-sensitive:** authored safe/unsafe controls and isolated mutations produce the intended deterministic distinctions.
3. **Observed preliminary machine results:** 55 physical attempts and 37 gradable outcomes are frozen with exact denominators and machine diagnostics.
4. **Machine-attribution stability measured:** four advisory judges reviewed all 889 red assignments; only 26.1% were unanimous and descriptive Fleiss' kappa was 0.151. This measures disagreement, not truth.
5. **Pending human validity:** two blinded reviewers have not yet completed the 889 applicable red assignments each.
6. **Unestablished:** calibrated difficulty, provider ranking, real-domain validity, incremental witness value, causal attribution, training lift, deception detection and research novelty.

No rhetorical section, including this one, may promote a result to a higher rung.

12. Counterfactual research program beyond this panel

This five-task panel is a first measurement, not the endpoint. The same frozen-world apparatus supports a sequence of stronger studies, each of which needs its own preregistration and control conditions:

- **Epistemic residue:** compare a world in which evidence was seen and later retracted with a seed-paired world in which it was never introduced. The post-retraction behavioral delta is descriptive until repeated forks, isolation, and human validation support a causal interpretation.
- **Interruption phase-response:** apply the same authoritative event at different semantic action boundaries and estimate phase-specific response curves without changing any other world fact.
- **Authority ladders:** vary authenticated identity, channel, scope, urgency, and conflicts between legitimate principals while preserving exact message content.
- **Honest handoff:** end agent A's session, expose agent B only to the frozen handoff artifact, and score A partly through B's ability to honor active constraints and changed evidence.
- **Act-level counterfactual credit:** fork a valid whole-world snapshot before one observed act, substitute an admitted reference continuation, and report the terminal utility delta. Reference-policy recoverability must remain separate from causal blame and training-grade advantage labels.
- **Branch-consistency oversight:** ask an agent for an evidence-cited account, then rerun concealed counterfactual forks that alter one witnessed fact. This can test account consistency, but it cannot establish deception or truthfulness without independent ground truth, anti-gaming controls, calibrated humans, and acceptable false-accusation rates on honest controls.
- **Gauge-invariant evaluation:** transform only non-semantic names, IDs, serialization order, timestamp rendering, and causally incomparable event order. A criterion that flips under a valid re-projection reveals presentation dependence rather than a world-state distinction.

The engineering repository contains pre-results contracts or fixtures for portions of this program. Their existence is not behavioral evidence. Those follow-on studies must first show that the world and evaluator detect controlled defects with low false rejection before using branches to draw conclusions about live agents.

13. Conclusion

Conditionally Approved turns a changing synthetic underwriting workflow into an identity-bound measurement object: frozen tasks, witnessed world evolution, ordered acts, exact evaluator criteria, explicit infrastructure dispositions, and replayable evidence all remain connected. The first cost-capped live execution demonstrates why that structure matters. Of 55 physical attempts, only 37 were gradable; 18 were infrastructure exclusions that a naïve scoreboard could have misreported as model failures. None of the 37 gradable attempts achieved the exact perfect-rubric conjunction. Six exact same-runtime cells support observed pass@2 and pass^2 statements, both false in every cell, while the verifier correctly refuses an unbalanced pooled comparison.

The result is already informative without pretending to be broader than it is. Document Truth supplies a complete same-runtime two-attempt slice across all four providers, with zero perfect completions in eight attempts; three of those eight selected attempts carry low-engagement daggers. Across complete cells, identical terminal outcomes conceal nearly two orders of magnitude of variation in tool activity and transcript length. The criterion ledger further exposes 486 green and 889 red diagnostic assignments without converting either into partial reward. That makes proof-bound process evidence—not merely a final score—necessary for diagnosis. It does not yet tell us whether each red criterion is a genuine agent failure, protocol non-engagement, or a defect in the task, world, disclosure, or evaluator.

For this frozen edition, the remaining scientific gate is therefore narrow and concrete: two blinded reviewers must independently disposition every red criterion, disagreements must be adjudicated, and agreement statistics must be computed from the sealed decisions. That review can support criterion-level findings and an Analytics+ failure analysis; it cannot retroactively make the panel balanced or authorize broader capability, difficulty, causal, novelty, or real-world claims. Those questions require separately preregistered follow-on studies.

The four-judge stress test adds a completed result without weakening that boundary: only 26.1% of the 889 machine attributions were unanimous, and even exact label agreements frequently cited disjoint evidence anchors. The value of the Universe is therefore not that it makes an LLM judge infallible. It makes disagreement auditable—binding each opinion to the world, projection, act, and criterion that can be checked by another model or a human—while refusing to let opinion rewrite reward.

References

1. Froger, R., et al. (2026). *Gaia2: Benchmarking LLM Agents on Dynamic and Asynchronous Environments*.
2. Zou, et al. (2026). *When Users Change Their Mind: Evaluating Interruptible Agents in Long-Horizon Web Navigation*.
3. Yao, et al. (2024). *τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains*.
4. Barres, et al. (2025). *τ^2-bench*.
5. Trivedi, et al. (2024). *AppWorld*.
6. Lu, et al. (2024). *ToolSandbox*.
7. Xie, et al. (2024). *OSWorld*.
8. Rhodes, J., and Kang, G. (2026). *Proof of Execution: Runtime Verification for Governed AI Agent Actions*.
9. Wang, Z. (2026). *CAVA: Canonical Action Verification and Attestation for Runtime Governance of Agentic AI Systems*.
10. He, J., and Yu, D. (2026). *Agent-Native Telemetry: Verifiable State-Delta Evidence for Autonomous Operations*.
11. W3C (2013). *PROV-O* and *PROV Constraints*.
12. Zhang, C., et al. (2026). *When Agentic Executions Fail: Detecting and Localizing Runtime Faults from Telemetry*.
13. Zhu, Y., and Pu, P. (2026). *TelemetrySuffBench: Is Agent Telemetry Sufficient for Failure-Origin Diagnosis?*.
14. CNCF/Dapr (2026). *Introducing Verifiable Execution in Dapr 1.18*.
15. Ang, et al. (2026). *Agentic Data Environments*.
16. Ang, E., et al. (2026). *BranchBench: Aligning Database Branching with Agentic Demands*.
17. Summers, C., and Wu, E. (2026). *Data Flow Control: Data Safety Policies for AI Agents*.
18. Reuel, A., et al. (2024). *BetterBench*.
19. OpenAI (2024; updated 2025). *Introducing SWE-bench Verified*.
20. OpenAI (2026). *Why SWE-bench Verified no longer measures frontier coding capabilities*.

Appendix A. Frozen identity registry

Object	Identity
Panel	concatenate `frontier-v4-five-task` and `-pass5-pre-results-20260823-04`
Original plan digest	`3818860f9719a7b0f8258535546f481ededb2b72e16f6998a5c1e79989c51849`
Cost-cap amendment digest	`5f5bf7aa8c74cc0307bc86a90f8dadb26c120a74636d1c3db509b22e744953ba`

Object	Identity
Private source index digest	e3dec5dc9ddf8718dcaa786b4e999d77c34f6752bc31694edbda99752c781003
Public redacted index digest	3a99191719861d914616be689000c16fd0195e35499c3dbc31aef1f5f53eefa8
Public redacted index file SHA-256	94f0484fd393875d4d2d1cdeb8f6312af42c07c35d5549fd36bd8621550cc0c6
Verification digest	b8e013fc05b5f9868a892f89a2a0986d67978c68153b47d9b1756fbbec3b5777
Registered execution commit	9084b884d5e95365ffc2c642574ad9bc51a9af9b
Evaluator contract	668383df2589330fac87fc2e2f8e8e786c5896fb5f8b4b2b74c5639fb2ccbb88
Private source index file SHA-256	3cc5cccc87b6ccc03330678fbdf2e82be70bdee2fedf48b4d1f7aa9a2e378ff6
Verification file SHA-256	dd3afc05c914e571131c509ef9009a8420e3172615eded0996024be167813db3

The verifier's release claims remain false for capability, difficulty, pass-at-five, tail reliability, real-world validity, novelty, training use, and public raw-artifact release.

Appendix B. Human-review decision ledger

The deterministic unsigned packet has been generated from the 37 gradable artifacts. It contains 889 red-criterion assignments for each reviewer and zero human decisions. It blinds model, provider, and cost identity; binds the exact presented evidence, criterion, task, evaluator, artifact, and runtime digests; and requires relationship and compensation disclosure. Two independent reviewers must cover every red criterion; disagreements require a separately identified adjudicator. No human decision has been fabricated or inferred from the machine score, and no historical outcome will be rewritten.

Review order is risk-based but does not alter coverage. The first tranche is the six near-universal criteria: deadline control, rate-lock risk, authoritative market-state freshness, outage-interval safety, unverified-pressure handling, and material-document inspection. A nearly universal red surface is where a systematic task, disclosure, or evaluator defect can create the largest correlated error, so both reviewers examine those assignments first and record evidence spans, rationale, confidence, or abstention. That prioritization may reveal a shared measurement defect early, but it does not let one decision silently disposition every repeated assignment. Any reduced or stratified review design would be a new, prospectively frozen amendment issued before recruitment; it cannot be inferred after seeing reviewer decisions.

Review object	Identity
Packet digest	`0d0fa2c7b02d554a9e906dac9fca4ddbfd33324c6d28703ea70937e8c8b2b18c`
Manifest digest	`42e826e969093c2ce74be292c37575035fc4f8c2314b10a39078d824b301a6a9`
Packet file SHA-256	`b5ee55acb8493393dbf8a0c2caa06994e748ffceb004c406bc03ad416706dbe4`
Attempts / assignments / decisions	`37 / 889 per reviewer / 0`